CS6P05ES Computing Research Project · Interim Viva

Multi-Class Brain Tumour MRI Classification

CNNs, Fine-Tuned Models and Ensemble Learning, with an Explainable-AI Uncertainty Signal

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Agenda

1. **Problem and aim**
   Why automated brain tumour classification
2. **Method and dataset**
   Three paradigms, one fair protocol
3. **Models built**
   CNN, pretrained models, feature extraction, ensemble
4. **Results so far**
   Interim accuracies and what they tell us
5. **Explainable AI plan**
   The novel EAA-IoU agreement signal
6. **System build and further work**
   Web application, timeline, next steps

Problem and Motivation

- MRI is central to brain tumour diagnosis, but reading scans manually is slow and varies between radiologists.
- Four categories must be separated: glioma, meningioma, no tumour, pituitary. Mistakes change the treatment path.
- Most published systems use a single model and give no signal of how confident or consistent that model is.
- Clinical need: an accurate second opinion that also flags the cases it is unsure about.

9,047

MRI scans · 4 classes · 2,063 held-out test images

SCREENSHOT PLACEHOLDER

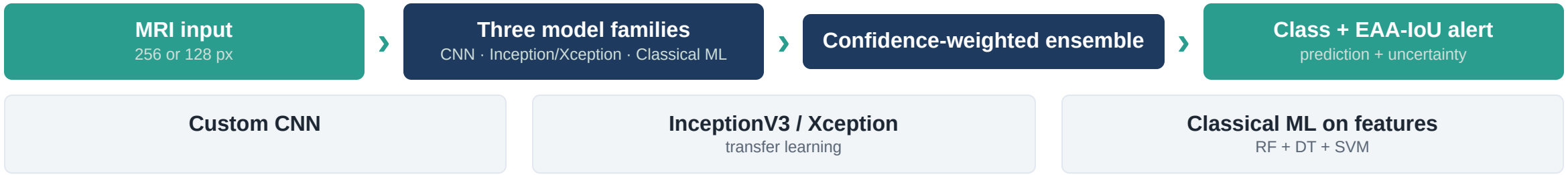
Four example MRI scans, one per class

Aim and Objectives

Aim. Design, optimise and compare CNNs, fine-tuned models and ensemble learning for multi-class brain tumour classification under identical conditions, and deploy it as a clinical web application with an explainable uncertainty signal.

- Evaluate CNN and classical methods for MRI feature extraction.
- Build a modular framework integrating CNNs, pretrained models and classical classifiers.
- Compare all models with one consistent metric set (accuracy, precision, recall, F1).
- Develop the EAA-IoU inter-model agreement signal from Grad-CAM, with no boundary labels.
- Deploy an interactive web application and document everything for reproducibility.

Methodology: One Framework, Three Paradigms

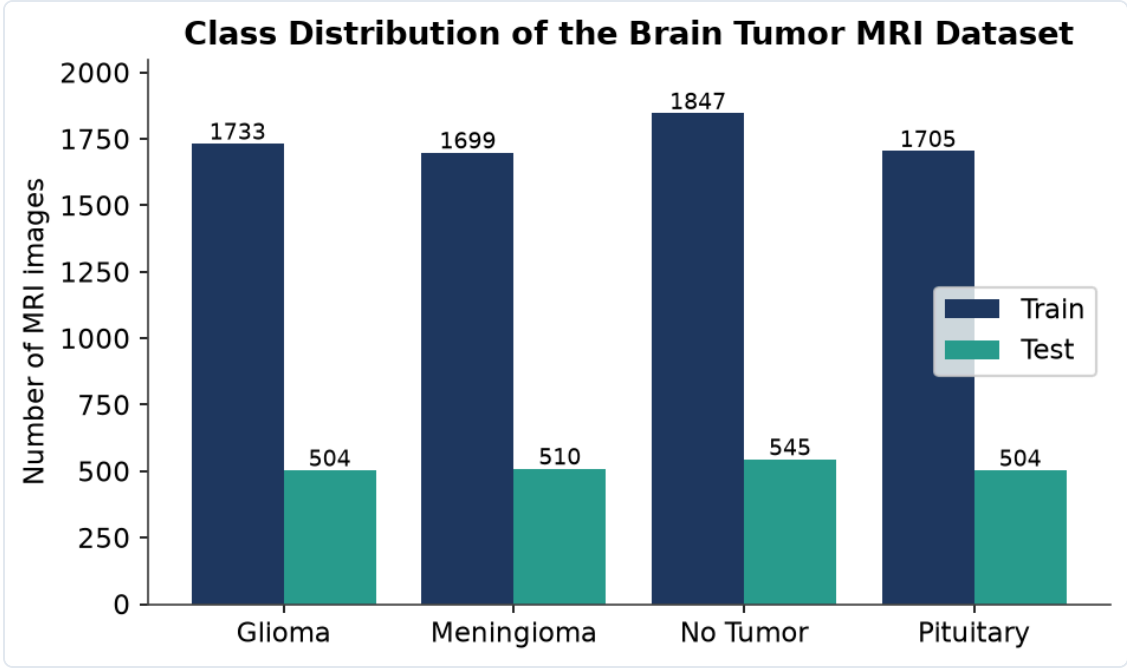


- Positivist, deductive, mixed-methods design: quantitative metrics now, clinician feedback later.
- Every model shares one data loader, one normalisation, one seed and the official test split, so the comparison is fair.

Dataset and Preprocessing

Class	Train	Test
Glioma	1,733	504
Meningioma	1,699	510
No Tumour	1,847	545
Pituitary	1,705	504
Total	**6,984**	**2,063**

- Resize to the model input size, normalise pixels to 0 to 1.
- Stratified split keeps class balance.
- Official test folder held out, used only for final scoring.



Class distribution across the train and test splits

What Is Built, What Remains

Completed

- Standardised, reproducible training framework
- Custom CNN trained and evaluated
- Classical ML baseline on raw pixels
- InceptionV3 transfer learning
- Classical ensembles on CNN / Inception features
- Models and result galleries hosted on Hugging Face

In progress and pending

- Xception fine-tuning
- Six-model integrated ensemble and confidence-weighted vote
- EAA-IoU explainability and validation
- Streamlit / FastAPI web application (scaffold built)
- Qualitative evaluation with clinicians

Model 1: Custom CNN

training/models/cnn.py

```
def build_cnn(input_size, feat_layer_name='dense'):
    model = models.Sequential([
        layers.Conv2D(32,(3,3),activation='relu',
            input_shape=(input_size,input_size,3)),
        layers.MaxPooling2D((2,2)),
        # three more Conv + MaxPool blocks ...
        layers.Flatten(),
        layers.Dense(256,activation='relu',
                     name=feat_layer_name),   # feature layer
        layers.Dropout(0.5),
        layers.Dense(4,activation='softmax'),
    ])
    model.compile(optimizer='adam',
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy'])
    return model
```

- Four convolution and pooling blocks learn tumour texture from coarse to fine.
- The named Dense(256) layer is kept as a feature point, reused later (feature extraction).
- Dropout 0.5 reduces over-fitting; softmax gives four class probabilities.
- **Trained 20 epochs, Adam. Test accuracy 91.18 percent**, the strongest single model.

Model 2: Pretrained Models (Transfer Learning)

training/models/transfer.py

```
def build_inception_standalone(size=256):
    base = InceptionV3(input_shape=(size,size,3),
        weights='imagenet', include_top=False)
    base.trainable = False        # freeze backbone
    x   = GlobalAveragePooling2D()(base.output)
    out = Dense(4, activation='softmax',
                name='dense')(x)  # new head
    return Model(base.input, out)
```

- InceptionV3 is pretrained on ImageNet, so it already detects edges, shapes and textures.
- include_top=False drops its 1000-class head; I add my own 4-class head.
- trainable=False freezes the backbone, so only the small head is learned. Fast, and works with limited medical data.
- **InceptionV3 84.63 percent.** Xception is the next model to train.

How We Extract the Dense Layer (Features)

training/evaluate.py

```
from tensorflow.keras.models import Model

def extract_features(model, layer_name, X):
    extractor = Model(
        inputs = model.input,
        outputs= model.get_layer(layer_name).output)
    return extractor.predict(X)   # (N, 256) vectors
```



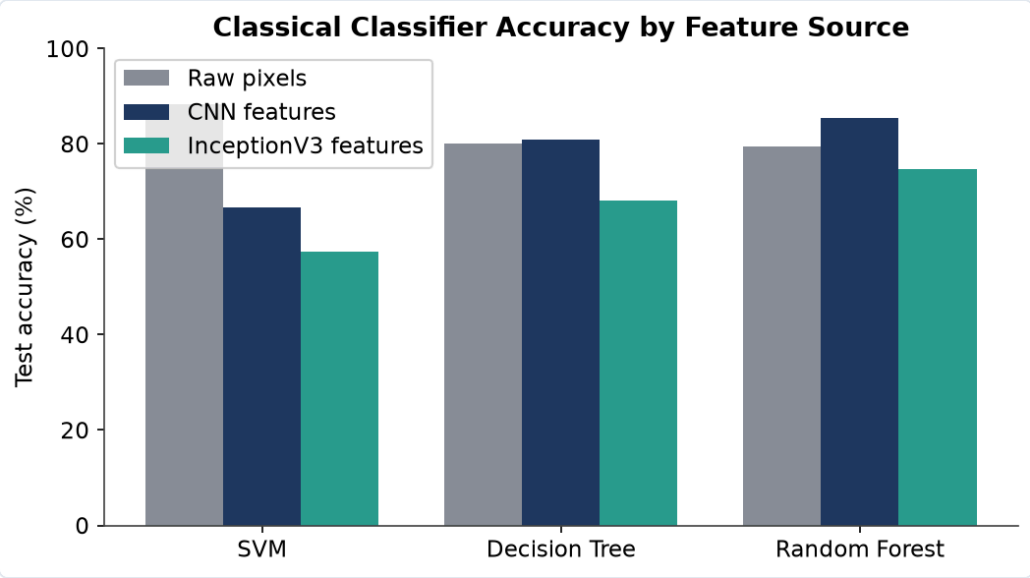
- A new Model is built that shares the same input but stops at the named layer.
- It outputs the 256-number activation vector, a compact summary of the image.
- No retraining: it reuses the trained weights and just reads an inner layer.
- CNN uses layer "dense", InceptionV3 "dense_2", Xception "dense_3".

Model 3: Classical Ensemble on Features

training/models/classical.py

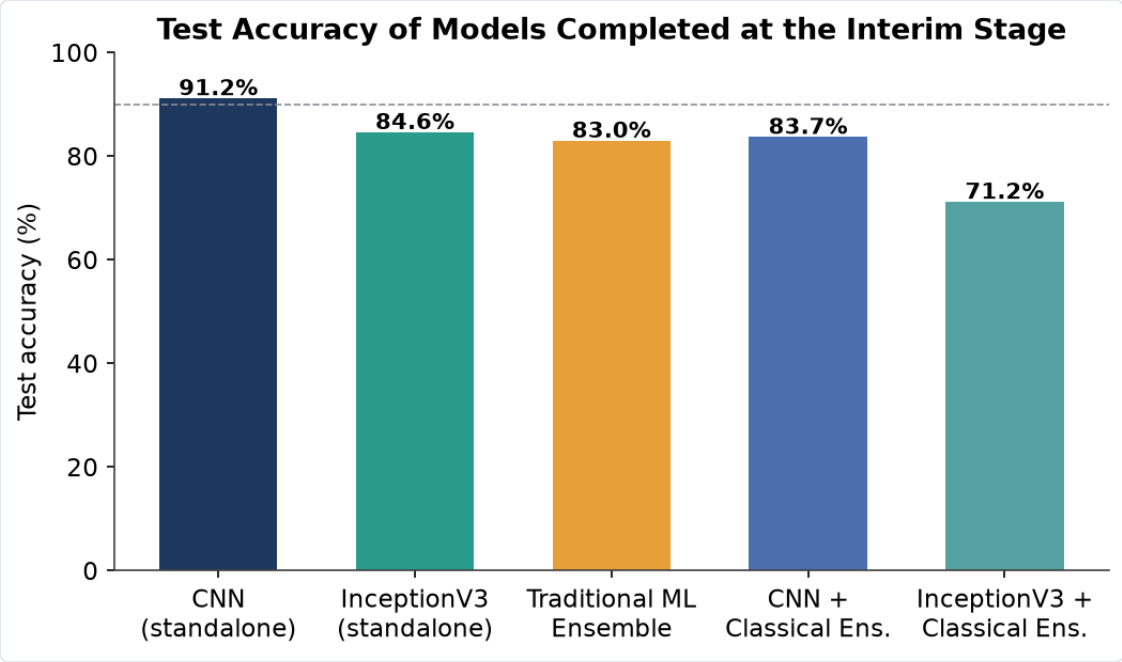
```
rf  = RandomForestClassifier(n_estimators=100)
dt  = DecisionTreeClassifier()
svm = SVC(probability=True)
ensemble = VotingClassifier(
    estimators=[('rf',rf),('dt',dt),('svm',svm)],
    voting='soft')          # average probabilities
ensemble.fit(train_feats, y_train)
```

- Soft voting averages the three classifiers probabilities, then picks the top class.
- Feature source matters: CNN features gave the best classical result (RF 85.45 percent).
- CNN feature ensemble 83.71 percent; Inception feature ensemble 71.20 percent.



Classical accuracy by feature source

Results So Far



Interim test accuracy

Reading the results

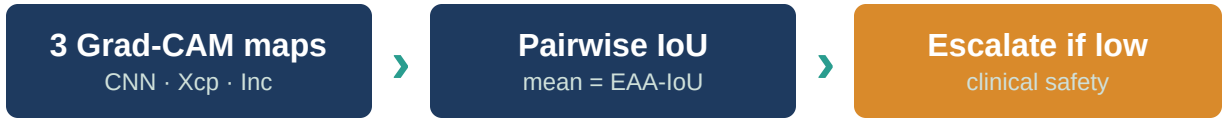
- Custom CNN is strongest so far at 91.18 percent.
- Raw-pixel SVM is a strong classical baseline at 88.26 percent.
- Glioma is the hardest class, most often confused with meningioma.
- This motivates the agreement signal: disagreement should concentrate on glioma.

Explainable AI: EAA-IoU Agreement (Novel)

webapp/xai_engine.py

```
def iou_score(a, b, t=0.5):
    a, b = a>=t, b>=t            # binarise heatmaps
    return (a & b).sum() / (a | b).sum()

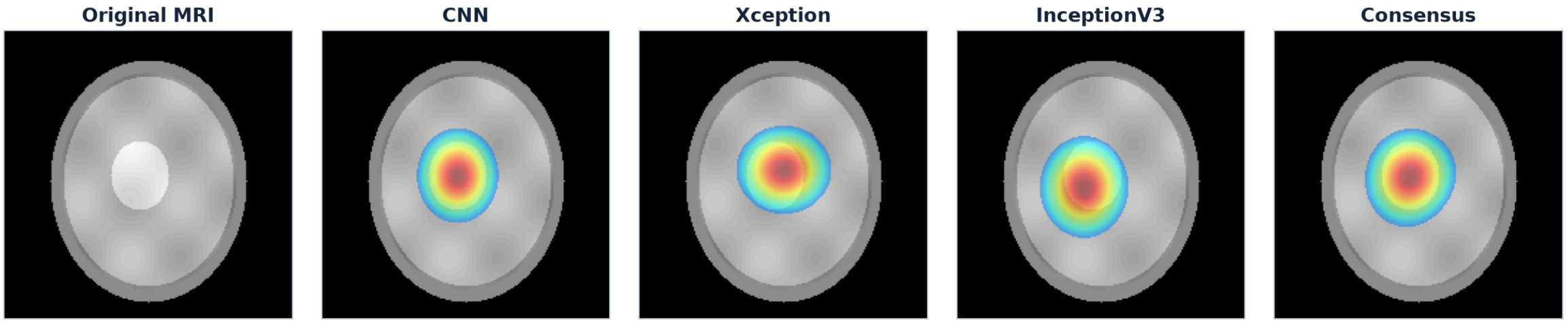
# per image: Grad-CAM from CNN, Xception, Inception
eaa = mean([iou(A,B), iou(A,C), iou(B,C)])
if eaa < 0.40:  flag_for_radiologist()
```



- Grad-CAM shows where each model looked. I compute the overlap (IoU) between the three maps.
- The mean pairwise IoU is one agreement score per image.
- Low agreement means the models disagree on the tumour location, so the case is flagged.
- **Novelty:** agreement is measured between models, needing no radiologist-drawn boundaries.

Grad-CAM Example and Agreement

Illustrative schematic of per-model Grad-CAM attention and the consensus map (not actual model output)



Illustrative schematic. Live heatmaps are generated per image in the web application.

- Each model produces its own Grad-CAM map showing the region it used to decide.
- Strong overlap between the three maps means the ensemble can be trusted.
- When the maps diverge, the EAA-IoU score drops and the case is flagged for review.

Worked example (illustrative)

`Pairwise IoU: 0.71, 0.69, 0.73`

`EAA-IoU = 0.71`

High agreement, high confidence

How the Software Is Built

Training modules
config · data · models · train

›

Hugging Face Hub
weights + results

›

Web app
FastAPI + React / Streamlit

- Training is modular Python: one config file fixes every hyperparameter and path.
- Trained models are versioned on Hugging Face; the app downloads them on first run.
- The app runs all models, builds Grad-CAM maps, computes the vote and the EAA-IoU alert.

SCREENSHOT PLACEHOLDER

Web application: upload, prediction, confidence, Grad-CAM panels, class probabilities

Further Work and Timeline

- Train Xception and complete the six-model integrated ensemble.
- Implement confidence-weighted voting for a real per-prediction confidence.
- Build and validate EAA-IoU on all test images, with class-wise analysis.
- Finish the web application and collect clinician feedback.

Phase	Status
1 Research and planning	Complete
2 Individual models	CNN, classical, Inception done
3 Ensemble framework	In progress
4 Evaluation	Partially done
5 Application	Scaffold built
6 Documentation	Ongoing

On schedule against the 16-week plan

Summary

- A fair, standardised comparison of CNN, pretrained and classical models is built and evaluated.
- Custom CNN leads at 91.18 percent; results so far are consistent and reproducible.
- The novel EAA-IoU agreement signal and the web application are the focus of the next phase.

Thank you. Questions and discussion.